

know. Of course, these companies use that data to their advantage, and yours as well, indirectly, by tailoring their services to suit your needs.

This is a form of machine intelligence, but calling it AI is stretching things a bit too far.

DESIRE

For a program to acquire sentience, it needs to have the desire to evolve and the means to do so. It must desire change and it must be creative enough to find ways to achieve that goal. Many scientists believed that the key to that was a sense of desire.

We like doing somethings because it makes us feel good, a sensation that's brought about by means of dopamine receptors in the brain. We do sometimes go too far and take drugs, but pleasure is a hugely important factor in driving our lives and it seems logical to design a computer or AI with the same system in mind. A reward system if you will.

You can't rig an AI to feel pleasure but you can tell it to achieve, say, a high score based on a certain set of rules. Sound familiar? That's exactly what DQN is, and the pitfalls are obvious now. If the only aim is to achieve a high score, it doesn't matter how you go about achieving it. As one scientist put it, if a robot's priority is to feed your kids, it might even go so far as to cook your pet in the oven. In its mind, the ultimate goal has been achieved. Thus, this system can never be the final solution.

If a machine is to learn, truly learn, it must understand and interpret actions and figure out why that behaviour is happening, what is it trying to maximise. Or so believes Dr. Stuart Russell (<http://bit.ly/1Qi5Bf5>) and he calls this technique, inverse reinforcement learning. In simpler terms, the machine is trying to understand our motivations. If this method actually works, and it certainly does look promising, an AI of the future will know why we're in a hurry or understand why we're excited for a new game release, or even that your pet is out of bounds are far as a food source is concerned. And this brings us neatly to the next great challenge facing AI: morality.

AN ETHICAL BEING?

What are morals? What is the difference between right and wrong? Hitler was a tyrant but at the time he rose to power,

his actions seemed entirely justified to the German public. Your self-driving car sees that an accident is inevitable, but does it run over your favourite dog that's standing in the middle of the road or that drunk beggar who is sprawled on the sidewalk?

One thing's for sure, if we're to create a super-being, it better have a sense of morality or we're doomed.

The problem with morals is that we ourselves don't know what exactly constitutes them. Each one of us has our own set of rules that we follow and we do our best to abide by them. Humanity in



Sentience is driven by desire

general might have a set of global rules, such as the fact that kids are more precious than adults, or that a human life is to be valued more than that of an animal, etc. We might be able to implant these basics on an AI, but it'll just have to learn the rest. But what if it learns from the "wrong" sort? Imbued with a sense of morality, will AI then evolve a sense of justice? Will it then pass judgement?

MIGHT IS RIGHT

Traditionally, it's always been war that's kick-started a new era of research and development and it's not much of a stretch to say that militaries around the world are spending a fair bit on AI-related research.

Right now, AI is only supporting humans in the performance of their duties (aircraft virtually fly themselves, especially military aircraft). The automation of

things such as Navigation, basic manoeuvres and reconnaissance will greatly lighten the load on soldiers. Modern targeting systems and friend or foe would be nowhere without silhouette recognition and ballistic (flight trajectories) algorithms and many defence systems, such as the Phalanx CIWS used on US ships, are almost completely automated to great effect. There are plans to automate reconnaissance and assault drones, the latter with the potential of being particularly deadly because the lack of a human pilot directly translates to lighter weight, much greater manoeuvrability and a level of decision making that would be unaffected by pilot skill or exhaustion.

Without a doubt, the future is drone warfare. But just as clearly, it'll be a long while before we see an AI entity with the inclination and the power to wield a military force to any sort of effect. Put another way, future wars will be fought by puppets on strings (drones), but it'll still be the puppet masters (us) who will start that war. At least for the time being.

SO WHERE DO WE STAND?

Our doom is nigh? Nay. AI is still too much in a nascent stage to even consider the possibility of its sentience anytime soon. In fact, the only other non-human creature (we're not even talking about programs) to ever ask an existential question was Alex the parrot (the bird, not the drone) who wanted to know what colour he was, and we can safely say that there is currently no AI in existence with even the rudimentary intelligence displayed by Alex. So far, the so-called AI of today is just a bunch of algorithms designed to interpret and respond to a very finite subset of data in extremely controlled and limiting environments. This AI can be lethal if programmed to kill and given control of a weapons platform, but even then, it won't really know what it was doing. We would still be the architects of our own destruction.

Will AI be a better strategist than the best generals, a better chef, a better painter, a singer? It's within the realms of possibility, but not yet. We're yet to see that one key event/discovery/algorithm that will transform simple software into true AI. We will know when that moment will come for that will be the day that we first create life. The day we play God. 